

Mechanics of the Preload Scanner Severing

How `<source media="(max-width: 600px)">` mechanically halts the browser's greedy network queue.

Standard `<img>` Pipeline

(Unconditional Greedy Fetch)

```

<!-- No media conditions present -->
```

🔍 Preload Scanner Thread

Scans the raw HTML byte stream solely for `src` strings. No viewport variables are queried.

⚡ Unconditional Push

Asset URL is instantly pushed to the Network Thread queue before DOM is even constructed.

🌐 Network Thread

GET /wide.jpg
(Maximum payload dispatched blindly)

The Baseline Flaw:

On mobile endpoints, this algorithm forces the download of a massive `wide.jpg` file, consuming bandwidth for pixels that will ultimately be suppressed by CSS later.

`<picture>` Art Direction Pipeline

(Conditional Fetch via Severing)

```
<picture>
  <source media="(max-width: 600px)"
    srcset="tight-crop.jpg">
  
</picture>
```

🔍 Preload Scanner Thread

Identifies `<source>` tag and extracts the `media` query string.

🛑 HALT: GREEDY FETCH SEVERED

✂️ The Mechanical Severing

The scanner evaluates the `media` query instantly against the browser's known environment viewport. It mechanically severs the greedy fetch queue, bypassing any dependency on the DOM or CSSOM.

The Mechanical Impact:

The `media` attribute acts as a logical circuit breaker. It mathematically ties the network request to the physical endpoint parameters, successfully severing the standard unconditional HTML parsing algorithm.

📏 Viewport Engine Evaluator

Is internal viewport width $\leq 600\text{px}$?
(Evaluated per Environment Viewport Metrics)

TRUE (Mobile)

FALSE (Desktop)

🌐 Network
GET /tight-crop.jpg

🌐 Network
GET /wide.jpg